§4 Exercises

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4.1. A 1-form on $\mathbb{R}^3$

Let ω be the 1-form $z\,\mathrm{d}x - \mathrm{d}z$ and let X be the vector field $y\partial/\partial x + x\partial/\partial y$ on $\mathbb{R}^3$. Compute $\omega(X)$ and $\mathrm{d}\omega$.

Proof. Beginning with the formula as derived in the text,

$$\begin{aligned}\omega(X) &= \sum b_i a^i \\ &= zy + 0 * x - 1 * 0 \\ &= zy.\end{aligned}$$

For the differential,

$$\begin{aligned}\mathrm{d}\omega &= \mathrm{d}(z) \wedge \mathrm{d}x + \mathrm{d}(-1)\,\mathrm{d}z \\ &= \mathrm{d}z \wedge \mathrm{d}x + 0 \\ &= \mathrm{d}z \wedge \mathrm{d}x.\end{aligned}$$

□

4.2. A 2-form on $\mathbb{R}^3$

At each point $p \in \mathbb{R}^3$, define a bilinear function ω_p on $T_p(\mathbb{R}^3)$ by

$$\omega_p(\mathbf{a}, \mathbf{b}) = \omega_p\left(\begin{bmatrix} a^1 \\ a^2 \\ a^3 \end{bmatrix}, \begin{bmatrix} b^1 \\ b^2 \\ b^3 \end{bmatrix}\right) = p^3 \det \begin{bmatrix} a^1 & b^1 \\ a^2 & b^2 \end{bmatrix},$$

for tangent vectors $\mathbf{a}, \mathbf{b} \in T_p(\mathbb{R}^3)$, where p^3 is the third component of $p = (p^1, p^2, p^3)$. Since ω_p is an alternating bilinear function on $T_p(\mathbb{R}^3)$, ω is a 2-form on $\mathbb{R}^3$. Write ω in terms of the standard basis $\mathrm{d}x^i \wedge \mathrm{d}x^j$ at each point.

Proof. Since,

$$\begin{aligned}\omega_p(\mathbf{a}, \mathbf{b}) &= p^3 \det \begin{bmatrix} a^1 & b^1 \\ a^2 & b^2 \end{bmatrix} \\ &= p^3(a^1 b^2 - a^2 b^1)\end{aligned}$$

and

$$\begin{aligned}(\mathrm{d}x^1 \wedge \mathrm{d}x^2)_p(\mathbf{a}, \mathbf{b}) &= (\mathrm{d}x^1_p \wedge \mathrm{d}x^2_p)(a, b) \\ &= \mathrm{d}x^1_p(a)\,\mathrm{d}x^2_p(b) - \mathrm{d}x^1_p(b)\,\mathrm{d}x^2_p(a) \\ &= a^1 b^2 - b^1 a^2,\end{aligned}$$

we have $\omega_p(\mathbf{a}, \mathbf{b}) = p^3(\mathrm{d}x^1 \wedge \mathrm{d}x^2)_p(\mathbf{a}, \mathbf{b})$. □

4.3. Exterior calculus

Suppose the standard coordinates on $\mathbb{R}^2$ are called r and θ (this $\mathbb{R}^2$ is the (r, θ)-plane, not the (x, y)-plane). If $x = r\cos\theta$ and $y = r\sin\theta$, calculate $\mathrm{d}x, \mathrm{d}y$, and $\mathrm{d}x \wedge \mathrm{d}y$ in terms of dr and $\mathrm{d}\theta$.

Proof.

$$\begin{aligned}\mathrm{d}x &= \mathrm{d}(r\cos\theta) \\ &= \frac{\partial(r\cos\theta)}{\partial r} dr + \frac{\partial(r\cos\theta)}{\partial\theta}\,\mathrm{d}\theta \\ &= \cos\theta dr - r\sin\theta\,\mathrm{d}\theta,\end{aligned}$$

$$\begin{aligned}\mathrm{d}y &= \mathrm{d}(r\sin\theta) \\ &= \frac{\partial(r\sin\theta)}{\partial r} dr + \frac{\partial(r\sin\theta)}{\partial\theta}\,\mathrm{d}\theta \\ &= \sin\theta dr + r\cos\theta\,\mathrm{d}\theta,\end{aligned}$$

$$\begin{aligned}\mathrm{d}x \wedge \mathrm{d}y &= (\cos\theta dr - r\sin\theta\,\mathrm{d}\theta) \wedge (\sin\theta dr + r\cos\theta\,\mathrm{d}\theta) \\ &= \cos\theta dr \wedge \sin\theta dr + \cos\theta dr \wedge r\cos\theta\,\mathrm{d}\theta - r\sin\theta\,\mathrm{d}\theta \wedge \sin\theta dr - r\sin\theta\,\mathrm{d}\theta \wedge r\cos\theta\,\mathrm{d}\theta \\ &= \cos\theta dr \wedge r\cos\theta\,\mathrm{d}\theta - r\sin\theta\,\mathrm{d}\theta \wedge \sin\theta dr \\ &= r\cos^2\theta dr \wedge \mathrm{d}\theta + r\sin^2\theta dr \wedge \mathrm{d}\theta \\ &= r(\cos^2\theta + \sin^2\theta) dr \wedge \mathrm{d}\theta \\ &= r dr \wedge \mathrm{d}\theta\end{aligned}$$

□

4.4. Exterior calculus

Suppose the standard coordinates on $\mathbb{R}^3$ are called ρ, ϕ, and θ. If $x = \rho\sin\phi\cos\theta$, $y = \rho\sin\phi\sin\theta$, and $z = \rho\cos\phi$, calculate $\mathrm{d}x$, $\mathrm{d}y$, $\mathrm{d}z$, and $\mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z$ in terms of $\mathrm{d}\rho$, $\mathrm{d}\phi$, and $\mathrm{d}\theta$.

Proof.

$$\begin{aligned}\mathrm{d}x &= \mathrm{d}(\rho\sin\phi\cos\theta) \\ &= \sin\phi\cos\theta\,\mathrm{d}\rho + \rho\cos\phi\cos\theta\,\mathrm{d}\phi - \rho\sin\phi\sin\theta\,\mathrm{d}\theta,\end{aligned}$$

$$
\begin{aligned}
\mathrm{d}y &= \mathrm{d}(\rho \sin\phi \sin\theta) \\
&= \sin\phi\sin\theta\,\mathrm{d}\rho + \rho\cos\phi\sin\theta\,\mathrm{d}\phi + \rho\sin\phi\cos\theta\,\mathrm{d}\theta,
\end{aligned}
$$

$$
\begin{aligned}
\mathrm{d}z &= \mathrm{d}(\rho\cos\phi) \\
&= \cos\phi\,\mathrm{d}\rho - \rho\sin\phi\,\mathrm{d}\phi,
\end{aligned}
$$

$$
\begin{aligned}
\mathrm{d}x\wedge\mathrm{d}y\wedge\mathrm{d}z = & -\sin\phi\cos\theta\,\mathrm{d}\rho\wedge\rho\sin\phi\cos\theta\,\mathrm{d}\theta\wedge\rho\sin\phi\,\mathrm{d}\phi \\
& + \rho\cos\phi\cos\theta\,\mathrm{d}\phi\wedge\rho\sin\phi\cos\theta\,\mathrm{d}\theta\wedge\cos\phi\,\mathrm{d}\rho \\
& + \rho\sin\phi\sin\theta\,\mathrm{d}\theta\wedge\sin\phi\sin\theta\,\mathrm{d}\rho\wedge\rho\sin\phi\,\mathrm{d}\phi \\
& - \rho\sin\phi\sin\theta\,\mathrm{d}\theta\wedge\rho\cos\phi\sin\theta\,\mathrm{d}\phi\wedge\cos\phi\,\mathrm{d}\rho \\
= & \,(\rho^2\sin^3\phi\cos^2\theta + \rho^2\cos^2\phi\cos^2\theta\sin\phi \\
& + \rho^2\sin^3\phi\sin^2\theta + \rho^2\sin\phi\sin^2\theta\cos^2\phi)\,\mathrm{d}\rho\wedge\mathrm{d}\phi\wedge\mathrm{d}\theta \\
= & \,\rho^2(\sin^3\phi\cos^2\theta + \cos^2\phi\cos^2\theta\sin\phi \\
& + \sin^3\phi\sin^2\theta + \sin\phi\sin^2\theta\cos^2\phi)\,\mathrm{d}\rho\wedge\mathrm{d}\phi\wedge\mathrm{d}\theta \\
= & \rho^2[\sin^3\phi(\cos^2\theta+\sin^2\theta) + \cos^2\phi\sin\phi(\cos^2\theta+\sin^2\theta)]\,\mathrm{d}\rho\wedge\mathrm{d}\phi\wedge\mathrm{d}\theta \\
= & \,\rho^2(\sin^3\phi + \cos^2\phi\sin\phi)\,\mathrm{d}\rho\wedge\mathrm{d}\phi\wedge\mathrm{d}\theta \\
= & \,\rho^2\sin\phi(\sin^2\phi+\cos^2\phi)\,\mathrm{d}\rho\wedge\mathrm{d}\phi\wedge\mathrm{d}\theta \\
= & \,\rho^2\sin\phi\;\mathrm{d}\rho\wedge\mathrm{d}\phi\wedge\mathrm{d}\theta
\end{aligned}
$$

□

4.5. Wedge product

Let α be a 1-form and β a 2-form on $\mathbb{R}^3$. Then

$$
\begin{aligned}
\alpha &= a_1\,\mathrm{d}x^1 + a_2\,\mathrm{d}x^2 + a_3\,\mathrm{d}x^3 \\
\beta &= b_1\,\mathrm{d}x^2\wedge\mathrm{d}x^3 + b_2\,\mathrm{d}x^3\wedge\mathrm{d}x^1 + b_3\,\mathrm{d}x^1\wedge\mathrm{d}x^2.
\end{aligned}
$$

Simplify the expression $\alpha\wedge\beta$ as much as possible.

Proof.

$$
\begin{aligned}
\alpha\wedge\beta &= a_1\,\mathrm{d}x^1\wedge b_1\,\mathrm{d}x^2\wedge\mathrm{d}x^3 + a_2\,\mathrm{d}x^2\wedge b_2\,\mathrm{d}x^3\wedge\mathrm{d}x^1 + a_3\,\mathrm{d}x^3\wedge b_3\,\mathrm{d}x^1\wedge\mathrm{d}x^2 \\
&= (a_1b_1 + a_2b_2 + a_3b_3)\,\mathrm{d}x^1\wedge\mathrm{d}x^2\wedge\mathrm{d}x^3.
\end{aligned}
$$

□

4.6. Wedge product and cross product

The correspondence between differential forms and vector fields on an open subset of $\mathbb{R}^3$ in Subsection 4.6 also makes sense pointwise. Let V be a vector space of dimension 3 with basis e_1, e_2, e_3, and dual basis $\alpha^1, \alpha^2, \alpha^3$. To a 1-covector $\alpha = a_1\alpha^1 + a_2\alpha^2 + a_3\alpha^3$ on V, we associate the vector $\mathbf{v}_\alpha = \langle a_1, a_2, a_3\rangle \in \mathbb{R}^3$. To the 2-covector

$$
\gamma = c_1\alpha^2\wedge\alpha^3 + c_2\alpha^3\wedge\alpha^1 + c_3\alpha^1\wedge\alpha^2
$$

on V, we associate the vector $\mathbf{v}_\gamma = \langle c_1, c_2, c_3\rangle \in \mathbb{R}^3$. Show that under this correspondence, the wedge product of 1-covectors corresponds to the cross product of vectors in $\mathbb{R}^3$: if $\alpha = a_1\alpha^1 + a_2\alpha^2 + a_3\alpha^3$ and $\beta = b_1\alpha^1 + b_2\alpha^2 + b_3\alpha^3$, then $\mathbf{v}_{\alpha\wedge\beta} = \mathbf{v}_\alpha \times \mathbf{v}_\beta$.

Proof. First, compute

$$\begin{aligned}
\alpha\wedge\beta &= a_1b_2\alpha^1\wedge\alpha^2 + a_1b_3\alpha^1\wedge\alpha^3 \\
&\quad + a_2b_1\alpha^2\wedge\alpha^1 + a_2b_3\alpha^2\wedge\alpha^3 \\
&\quad + a_3b_1\alpha^3\wedge\alpha^1 + a_3b_2\alpha^3\wedge\alpha^2 \\
&= (a_2b_3 - a_3b_2)\alpha^2\wedge\alpha^3 \\
&\quad + (a_3b_1 - a_1b_3)\alpha^3\wedge\alpha^1 \\
&\quad + (a_1b_2 - a_2b_1)\alpha^1\wedge\alpha^2,
\end{aligned}$$

so $\mathbf{v}_{\alpha\wedge\beta} = \langle a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1\rangle$, which is precisely the definition of $\langle a_1, a_2, a_3\rangle \times \langle b_1, b_2, b_3\rangle = \mathbf{v}_\alpha \times \mathbf{v}_\beta$. □

4.7. Commutator of derivations and antiderivations

Let $A = \oplus_{k=-\infty}^{\infty} A^k$ be a graded algebra over a field K with $A^k = 0$ for $k < 0$. Let m be an integer. A *superderivation of A of degree m* is a K-linear map $D : A \to A$ such that for all k, $D(A^k) \subset A^{k+m}$ and for all $a \in A^k$ and $b \in A^\ell$,

$$D(ab) = D(a)b + (-1)^{km}a(Db).$$

If D_1 and D_2 are two superderivations of A of respective degrees m_1 and m_2, define their *commutator* to be

$$[D_1, D_2] = D_1 \circ D_2 - (-1)^{m_1m_2}D_2 \circ D_1.$$

Show that $[D_1, D_2]$ is a superderivation of degree $m_1 + m_2$. (A superderivation is said to be *even* or *odd* depending on the parity of its degree. An even superderivation is a derivation; an odd superderivation is an antiderivation.)

Proof. Let D_1 and D_2 be superderivations of respective degrees m_1 and m_2. First, we show that $[D_1, D_2]$ is K-linear. Given $a, b \in A$,

$$\begin{aligned}
[D_1, D_2](a+b) &= D_1\circ D_2(a+b) - (-1)^{m_1m_2}D_2\circ D_1(a+b) \\
&= D_1(D_2(a) + D_2(b)) - (-1)^{m_1m_2}D_2(D_1(a) + D_1(b)) \\
&= D_1(D_2(a)) + D_1(D_2(b)) - (-1)^{m_1m_2}D_2(D_1(a)) - (-1)^{m_1m_2}D_2(D_1(b)) \\
&= D_1(D_2(a)) - (-1)^{m_1m_2}D_2(D_1(a)) + D_1(D_2(b)) - (-1)^{m_1m_2}D_2(D_1(b)) \\
&= [D_1, D_2](a) + [D_1, D_2](b),
\end{aligned}$$

and for $r \in R$,

$$\begin{aligned}
[D_1, D_2](ra) &= D_1(D_2(ra)) - (-1)^{m_1m_2}D_2(D_1(ra)) \\
&= rD_1(D_2(a)) - r(-1)^{m_1m_2}D_2(D_1(a)) \\
&= r(D_1(D_2(a)) - (-1)^{m_1m_2}D_2(D_1(a))) \\
&= r[D_1, D_2](a).
\end{aligned}$$

Now suppose $a \in A^k$ and $b \in A^\ell$. First, observe that

$$[D_1, D_2](a) = D_1(D_2(a)) - (-1)^{m_1 m_2} D_2(D_1(a)),$$

and $D_1(D_2(a)), D_2(D_1(a)) \in A^{k+m_1+m_2}$. Thus, being a linear combination of these two, $[D_1, D_2](a) \in A^{k+m_1+m_2}$. Finally,

$$\begin{aligned}
[D_1, D_2](ab) &= D_1(D_2(ab)) - (-1)^{m_1 m_2} D_2(D_1(ab)) \\
&= D_1(D_2(a)b) + (-1)^{km_2} D_1(aD_2(b)) \\
&\quad - (-1)^{m_1 m_2} D_2(D_1(a)b) - (-1)^{m_1 m_2 + km_1} D_2(aD_1(b)) \\
&= D_1 \circ D_2(a)b + (-1)^{km_1 + m_2 m_1} D_2(a) D_1(b) \\
&\quad + (-1)^{km_2} D_1(a) D_2(b) + (-1)^{km_2 + km_1} aD_1 \circ D_2(b) \\
&\quad - (-1)^{m_1 m_2} D_2 \circ D_1(a)b - (-1)^{m_1 m_2 + km_2} D_1(a) D_2(b) \\
&\quad - (-1)^{m_1 m_2 + km_1} D_2(a) D_1(b) - (-1)^{m_1 m_2 + km_1 + km_2} aD_2 \circ D_1(b) \\
&= \left[D_1 \circ D_2(a) - (-1)^{m_1 m_2} D_2 \circ D_1(a)\right] b \\
&\quad + (-1)^{k(m_1+m_2)} a \left[D_1 \circ D_2(b) - (-1)^{m_1 m_2} D_2 \circ D_1(b)\right] \\
&= [D_1, D_2](a)b + (-1)^{k(m_1+m_2)} a[D_1, D_2](b)
\end{aligned}$$

□